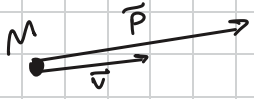
momentum

$$\vec{F} = m\vec{a} = m \cdot \frac{d\vec{v}}{dt}$$

if m is constant, $m\frac{d\vec{v}}{dt} = \frac{d}{dt}(m\vec{v})$

$$\Rightarrow \vec{F} = \frac{d}{dt}(m\vec{v})$$

def $\boxed{\vec{p} = m\vec{v}}$ momentum of the particle



$$\Rightarrow \boxed{\vec{F} = \frac{d\vec{p}}{dt}}$$

Alternative form of Newtons second Law

in the case when $\vec{F} = 0 \Rightarrow \frac{d\vec{p}}{dt} = 0$

$\Rightarrow \vec{p}$ constant

$\langle p \rangle = kg \cdot m/s$

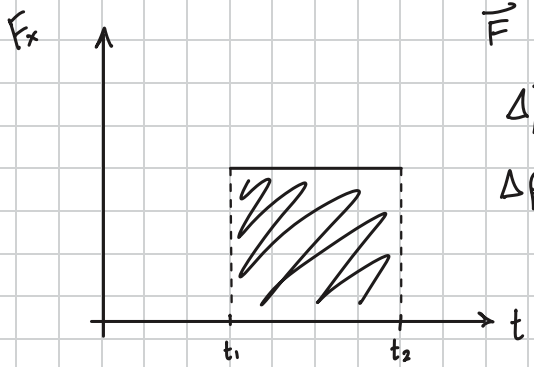
if $\vec{p}$ constant $\Rightarrow$ p_x, p_y, p_z Constant

$$\vec{F} = \frac{d\vec{p}}{dt} \Rightarrow F_x = \frac{dp_x}{dt}, \; F_y = \frac{dp_y}{dt}, \; F_z = \frac{dp_z}{dt}$$

$$\vec{F} = \frac{d\vec{p}}{dt} \quad \Rightarrow \quad \vec{p} = \int \vec{F}\, dt$$

by components

$$p_x = \int F_x\, dt, \quad p_y = \int F_y\, dt, \quad p_z = \int F_z\, dt$$



$\vec{F}$ = constant

$\Delta\vec{p} = \vec{F} \cdot \Delta t$

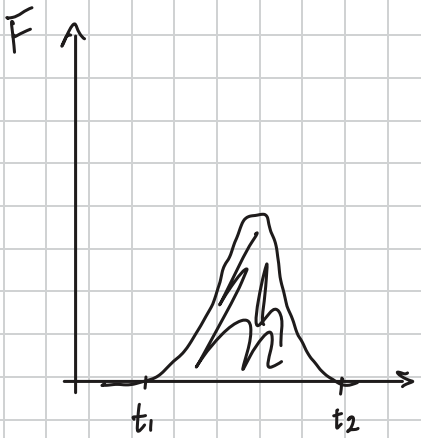
$\Delta p_x = F_x \Delta t$

$\vec{F} \cdot \Delta t = \vec{J} \rightarrow$ the impulse of $\vec{F}$

the impulse of the force equals the change in momentum of the object on which the force is acting

$$\vec{J} = \Delta \vec{p}$$

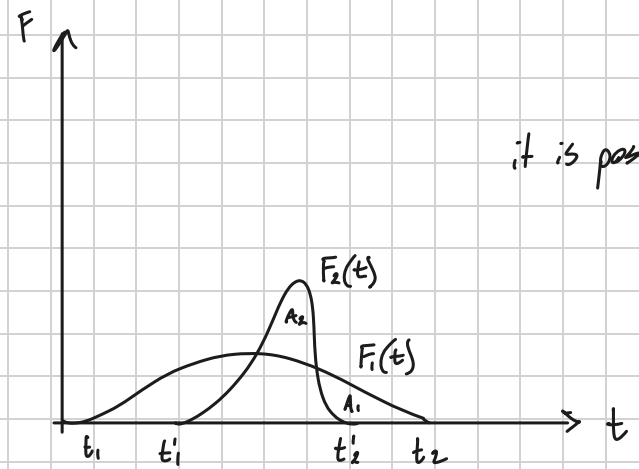
what if F isn't constant



$$J_{F_{t_1 \to t_2}} = \int_{t_1}^{t_2} F(t)\,dt$$

= Area under graph of F(x)

$$J_{t_1 \to t_2} = \int_{t_1}^{t_2} F(t)\,dt = F_{ave} \cdot (t_2 - t_1)$$



it is possible that $A_1 = A_2$

$$\int_{t_1}^{t_2} F_1(t)\,dt = \int_{t_1'}^{t_2'} F_2(t)\,dt \Rightarrow \quad J_1 = J_2$$

=> these two forces would generate the same momentum change on an object

Ex

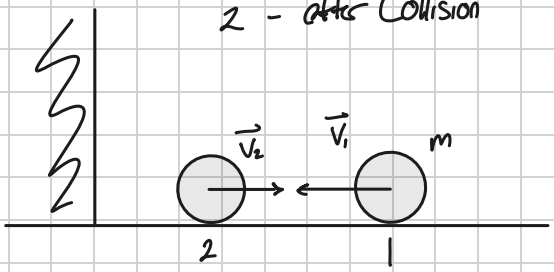
$m = 0.40\,kg$

1 - before

2 - after Collision

$|V_1| = 30\,m/s$

$|V_2| = 20\,m/s$



What is the impulse of the force of interaction with the wall during the collision?

$$J = \int_{t_1}^{t_2} F(t)\,dt$$

but $J = \Delta p = p_2 - p_1 = mV_2 - mV_1$

$V_1 = -30\,m/s$

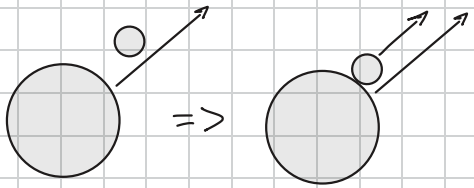
$V_2 = 20\,m/s$

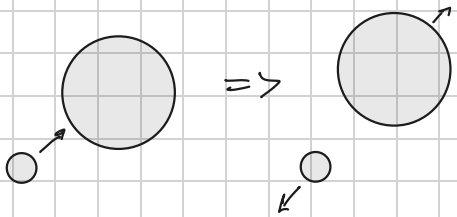
$J = 0.40\,kg \cdot 20\,m/s - 0.40\,kg \cdot (-30\,m/s)$

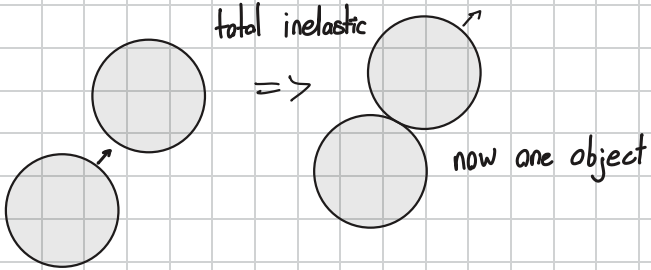
$= 0.40\,kg \cdot 50\,m/s$

$= 20\,kg \cdot m/s$

Collisions

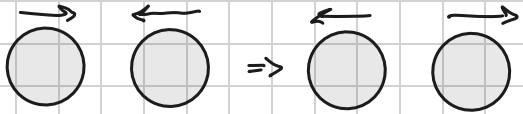






elastic collisions

- the total kinetic energy of the system is the same after the collision



Example

Mass A collides with mass B pure elastic

find final A and B velocity and initial velocity of A

Momentum is Always conserved

$$M_A V_{A_i} + \cancel{M_B V_{B_i}}^{0} = M_A V_{A_f} + M_B V_{B_f}$$

Solve for V_{A_f}

$$V_{A_f} = \left(\frac{M_A - M_B}{M_A + M_B}\right) V_{A_i} \quad ?$$